err.Error() to HTTP response enumeration (PX-F18, §11.4.276(C) class enumeration), rework 3 (2026-11-13)

Field	Value
Date	2026-11-13
Roadmap item	PX-F18 (CAT-PREEXIST-3)
Review base	main @ 9ed6577b (the base the NO-GO review of 2026-11-12 read)
Current base	main @ 713a4f380 (every "HEAD" below)
Scope	every non-test .go file under catalog-api (all packages, `main.go` included) plus the other Go modules of the repository that serve HTTP (OCU-CUDA-Sidecar; `scripts/test-infra/nas_clients` scanned: 0 hits)
Status	enumeration only; no source file changed. Replaces the doc the review rejected (it was never landed)
Corrections 2026-11-14	PXF18-FU1 to FU5 and FU7 to FU11 applied (FU6, the challenges/results admin guard, is handled elsewhere and is NOT touched here). Every corrected claim was re-measured at main @ 9e624f09f; where a claim differs between 713a4f380 and 9e624f09f both are stated. Scan numbers are still at the two bases above. See "Corrections of 2026-11-14"
Verdict labels	PROVISIONAL, review owed. Nothing here is a review GO

Why this is a rework

The NO-GO review (docs/FOLLOW_UPS_NEXT_PHASE.md, "PX-F18 enumeration doc review: NO-GO") found one unlisted class and five defects of the instrument and the wording. Each is closed in this document as follows (evidence is in the sections named).

Review finding	Closure	Where
F1: `utils.SendErrorResponse(c,status,msg,err)` puts `err.Error()` into `details`; 109 call sites in 11 files (media_entity 35, collection 15, comic_pages 14, stats 10, media 10, search 6, recommendation 6, challenge 5, browse 5, media_browse 2, middleware/auth.go 1; auth.go:167 sends JWT parse text on 401)	The scanner now matches a bare `err` argument. At the review base it reproduces the review's numbers exactly: 109 bare-err call sites in the same 11 files with the same per-file counts, plus one more SendErrorResponse site that passes `err.Error()` as the *message* (collection_handler.go:335). The class is enumerated per site (rows B1..B226) and the mechanism is verified: base `utils/response.go:31` has `response.Details = err.Error()`; HEAD has no assignment to `Details` anywhere in `utils/`	"Class 1" and the per-site table
F2: internal/metrics/health.go:157 (unauthenticated /health/deep leaked the DB ping error); monitoring.go:251 (unwired)	Both are rows H1, H2. They are struct/map-literal flows that a sink-call scanner cannot see, so a second instrument (every zero-argument `.Error()` call, any context) was added. health.go:157 is closed at HEAD (fixed text, error logged). monitoring.go:251 is still present at HEAD and is unwired (no non-test importer of `catalogizer/internal/monitoring`)	"Class 2", rows H1/H2, "HEAD residual" R1
F3: auth rows should be LEAK	Reclassified. At the review base `AuthService.Login` returns `fmt.Errorf("failed to get user: %w", err)` and similar chains (auth_service.go:105,147,170,177,183,189) and the handler sends `err.Error()` on 401, so the rows are LEAK, with authored sentinels (locked, disabled, pending) mixed in. The 20 former UNKNOWN rows were each resolved by reading the producer (rule list in "Classification rules")	per-site table, column Class
F4: middleware rows are latent	Reclassified LATENT (rows with `internal/middleware/middleware.go`). ErrorHandler is wired (`main.go:1038` at base, `:1045` at HEAD), but its two leaking branches need gin `ErrorTypeBind` or `ErrorTypePublic` errors, and no non-test code produces either (grep for `.Bind(`, `BindJSON`, `MustBindWith`, `AbortWithError(`, `ErrorTypePublic`: 0 hits outside the middleware file; `ShouldBind*` does not append to `c.Errors`). Third-party middleware that might append such errors was not enumerated: UNCONFIRMED	per-site table
F5: limits text wrong	The old text said "no such helper found by name search"; SendErrorResponse is exactly such a helper and 109 sites used it. "Limits of the instruments" is rewritten from what was measured	last section

Review finding	Closure	Where
F6: tool/main.go errSite misses bare `err` arguments	Fixed in instrument 1 (kind BARE). The control needle below fails on the old logic (bare err not reported) and passes on the new	"Control needles"

Round-2 review findings (NO-GO on rework 2), closure in this rework

Finding	Closure	Where
N1: "class complete / no unclosed LEAK" false: class excluded state persisted and later served; sink scan missed an err inside a helper in a sink (fixture f) and a non-err-named variable in %v (fixture g)	Class redefined as error-derived text that can reach a client by ANY path. Instrument 1 gained kind `H:` (any call with an err-like identifier argument inside a sink argument) and err-like detection by type/nil-comparison; reviewer fixtures f and g are now found. New instrument 3 (flow scan) lists error-derived text stored in struct fields, map/kv entries, channels and call arguments. New section "Class 4" traces each stored field to its writer, route, gate and scrub. The "no LEAK row remains unclosed" sentence is withdrawn	Class definition, Instruments, Class 4, Control needles
N1 specific rows (GetStorageRootStatus, ListStoragePath, SyncSession.ErrorMessage, ScanStatus.Reason + event bus, scrubSecrets and sync_service.go:1057 census rows)	Rows S1 to S5 in Class 4; the census sentence "none reach a response" is corrected	Class 4, Census
N2: unrouted net/http variants counted as live	B7 to B10, B12 to B14, B80 to B83, B213, B214 reclassified LATENT with the routing evidence; totals recounted	Per-site table, Review-base totals
N3: validationText/paramRequiredText rows wrongly "CLOSED"; claim "0 SendErrorResponse sites carry error text"	HEAD status of VALIDATION rows reworded to "NOT CLOSED"; collection_handler.go:335,351, playlist_handler.go:95,188, service_handlers.go:258,298 and input_validation.go:403 added to R4; the claim is replaced	R4, Class 1
N4: configuration_service.go:430,443, localization_service.go:849-904, recovery/retry. go:401, webdav_client.go:340-502	Traced in Class 4 rows S6 to S9 (S10 to S12 are census and flow-scan rows added in rework 3, all SAFE; corrected 2026-11-14, PXF18-FU10)	Class 4
N5: B2 reason, B3 "(unresolved in review base)", ErrRecommendationsNotImplemented	B2 now VALIDATION with the producer cited; B3 reason cites auth_service.go:132; the sentinel text is noted in Class 1	Per-site table, Class 1

Instruments

Three Go AST programs (go/parser, go/ast), built in a scratch directory; the sources are committed next to this doc as `docs/qa/err_error_class_scan/astscan_sinks.go.txt`, `all_error_calls.go.txt` and `flows_assign.go.txt` (`.txt` so no Go tool picks them up). Instruments are numbered 1 (sink scan), 2 (census), 3 (flow scan); the order of the items below was 0, 1, 3, 2 before the 2026-11-14 correction (PXF18-FU10). They were run on a clean clone at both bases, with each tree checked out without recursive submodule init.

- **Rework 3 changes to the sink scan** (not an instrument). Err-like identifiers are now found by type (`error` in parameters, results and `var` declarations) or by comparison with nil, so a variable named `failure` or `cause` counts. Kind `H:` reports any non-`fmt` call that takes an err-like identifier inside a sink argument. Kind `F:` matches any err-like identifier argument of `fmt.Sprint*`.

1. **Sink scan.** For every non-test function it finds calls whose callee name looks like a response sink (names containing json, abortwith, string, write, send, respond, errorresponse, badrequest, servererror, encode, `http.Error`, html, data, emit, broadcast, notify, plus `c.Error`/`AbortWithError`); logger, `fmt.*`, `errors.*`, `strings.*` callees are excluded. An argument is reported when its subtree holds (E) a zero-argument `.Error()` call, (F) `fmt.Sprint*`/`Errorf` with an err-named identifier argument, (T) an identifier that the same function assigned from such an expression (taint, flow-insensitive), or (BARE) an err-named identifier passed as an argument.
2. **Census.** Every zero-argument `.Error()` call in every non-test file, with its syntactic context (argument of a named call, assignment, return, composite literal / key-value, branch, other). It catches flows instrument 1 cannot see (struct literals returned to a handler, helpers with unusual names).

3. **Flow scan** (`docs/qa/err_error_class_scan/flows_assign.go.txt`, skips `challenges/`). Lists error-derived text (`err.Error()`, `%v` of an err-like identifier, a variable assigned from either) assigned to a struct field, an index or map key, a key-value pair, a channel send or a call argument, with file, line, function and kind. Its rows are leads for Class 4, not verdicts.

Control needles (§11.4.201(7)(b), §11.4.273)

- Known positive on the review base: `handlers/error_reporting_handler.go:85` (`http.Error(w, err.Error(), 500)`): found, 1 row, instrument 1. On HEAD that file has 0 rows (closed by CAT-PXF23-R5B).
- Synthetic fixture with four functions (bare err to `SendErrorResponse`; `fmt.Sprintf` into a variable then `c.JSON`; direct `err.Error()` in a map; `err.Error()` only in a log call and an == test): instrument 1 reported exactly the first three (kinds BARE, TEXT-via-taint, TEXT) and nothing for the fourth; instrument 2 reported the direct call (context kv), the log argument (`arg-of:log.Printf`) and the comparison (`branch`).
- Reviewer fixtures (round 2): `fixF` (`c.JSON(500, gin.H{"e": scrub(err)})`, helper nested in a sink) and `fixG` (`fmt.Sprintf("failed: %v", failure)` where `failure` is a non-err-named `error`) are both reported by the rebuilt instrument 1 (kinds `H:` and `F:`); good1 to good3 are still reported and bad1 is still silent. The previous build reported neither fixF nor fixG.
- Re-run of instrument 1 on a fresh checkout of each base reproduces the stored outputs: raw 432 rows (review base) and 371 rows (HEAD); 310 and 250 after removing the `strings.`, `sort.`, `bytes.` callee noise. New HEAD rows of kind `H:` are the helpers listed in Class 3 and R4 plus nil-comparison noise (`len(playlists)`, `[]string(p)` and similar, read and discarded as false positives).
- Fabricated token `ZZNEEDLE_NOPE`: 0 hits in all output files (instruments 1, 2 and 3, both bases).
- Cross-check of instrument 2 against `grep -rE '\.Error\(\)'` (non-test .go, catalog-api): review base AST 333 calls vs 329 grep lines; HEAD AST 255 vs 254. Line-set comparison (LC_ALL=C): AST-only lines 0 at both bases; grep-only lines are 1 at the review base and 4 at HEAD (universal_scanner.go:188 in both), and every one was read: they are comments that quote `err.Error()` (universal_scanner.go:188, auth_handler.go:36, client_errors.go:67, utils/response.go:31). The AST count exceeds the grep line count because several lines hold more than one call (5 such lines at each base).
- Taint is flow-insensitive: on HEAD `handlers/user_handler.go:122` is reported as `T:msg` because `msg` is assigned from `err.Error()` later in the function (line 168); at line 122 `msg` comes from `authClientMessage` (authored text). Read and confirmed a false positive; it is not counted below.

Class definition

A site is in the class when text derived from an `error` value can reach a client (HTTP, WebSocket or SSE response body or header) by ANY path. Routes (rework 3 widens this; the earlier definition excluded server-side state that is persisted and later served): (1) `err.Error()` or `%v/%w/%s` of an error inside a response call; (2) a bare error passed to a helper that serialises it; (3) error text stored in a struct or map that a handler later serialises; (4) error text kept in a variable and used in a response; (5) error text returned from a helper into a response; (6) a helper called inside a sink argument (`gin.H{"e": scrub(err)}`); (7) `%v/%s` of an error-typed identifier without `.Error()`, whatever the identifier is called; (8) error text stored in a struct field, map, database row, session or status object and served later by another route (Class 4); (9) error text published on the event bus and forwarded to WebSocket or SSE clients. Sites that only log or branch on text are outside the class; sites that record text in server-side state are IN the class when a route serves it, and are traced in Class 4.

Classes used in the tables: **LEAK** (internal detail can reach the client: SQL/driver text, wrapped `%w` chains, paths, upstream or JWT parse text; "can include", not a demonstrated leak for every row: service and repository strings were read only where stated); **VALIDATION** (authored text that tells the caller which input to fix, or a sentinel's own text; may echo the caller's own input); **LATENT** (leaking code that no non-test caller can reach); **SAFE** (the err value does not reach the client: log-only, fixed-text helper, or a non-client sink); **CLOSED-BY-HELPER** and **CLOSED** are HEAD statuses, not classes.

Classification rules (how the 20 old UNKNOWN rows were resolved)

Rule 1: a 4xx/5xx text produced by a service chain that contains `fmt.Errorf("...: %w", repoErr)` is LEAK (configuration GetWizardProgress, error_reporting GetErrorReport/GetCrashReport, auth Login/Refresh/ChangePassword via auth_service.go wraps, localization Import/Edit). Rule 2: text from a package-level sentinel or a `smartrules.ValidationError` is VALIDATION (playlist, service_handlers, conversion). Rule 3: `ErrLocalRootNotAllowed` text from `ResolveLocalRoot`/`CheckLocalRoot` is LEAK (it embeds `cannot resolve path: <OS error>`, local_policy.go:54). Rule 4: a binder (ShouldBindJSON) error on 400 is VALIDATION (mild: names Go struct fields). Rule 5: any other text on a 5xx or a variable status is LEAK.

Class 1: SendErrorResponse family (the review's unlisted class)

`utils.SendErrorResponse(c, status, message, err)` at the review base (`utils/response.go:25-39`): if `err != nil`, `response.Details = err.Error()`, for every status. Bare-err call sites by file (review base):

File	Sites
catalog-api/handlers/media_entity_handler.go	35
catalog-api/handlers/collection_handler.go	15
catalog-api/handlers/comic_pages_handler.go	14
catalog-api/handlers/media_handler.go	10
catalog-api/handlers/stats.go	10
catalog-api/handlers/recommendation_handler.go	6
catalog-api/handlers/search.go	6
catalog-api/handlers/browse.go	5
catalog-api/handlers/challenge.go	5
catalog-api/handlers/media_browse_handler.go	2
catalog-api/middleware/auth.go	1
Total	**109** in 11 files

The review base has 167 SendErrorResponse call sites in non-test code: 109 pass a bare err variable (the table above; `nerr`/`rerr` at comic_pages_handler.go:582,592 are included), 1 passes `err.Error()` as the message (collection_handler.go:335), and 57 pass `nil` or a sentinel (`ErrRecommendationsNotImplemented`, recommendation_handler.go:115,152). The 57 carry no error text and are not rows.

HEAD: `SendErrorResponse` logs err (status, request id) and never sets `Details`; `ErrorResponse.Details` is no longer assigned in `utils/` (grep `Details =`: base 1 hit at response.go:31, HEAD 0), with `TestSendErrorResponse_DoesNotLeakInternalErrorText` and `..._HeadersDoNotLeakErrorText` in `utils/response_test.go`. All 109 rows are therefore CLOSED-BY-HELPER. Residual risk: the helper's `message` argument. At HEAD the `message` argument carries error-derived text at 2 SendErrorResponse sites: collection_handler.go:335 and :351, both `validationText(err)` (authored `ValidationError.Msg`, VALIDATION, listed in R4). The earlier sentence "0 TEXT rows" described only kinds `E:`/`F:`/`T:` and missed the helper kind `H:`; it is withdrawn. The 57 sites that pass `nil` or a sentinel include `ErrRecommendationsNotImplemented` (recommendation_handler.go:115,152): at the review base its own text "recommendation source not implemented" went into `details`; it is authored text, not internal detail (not counted as a row).

Class 2: error text in struct/map literals and indirect flows

Found only by instrument 2 or by taint, not by the sink scan:

Row	File:line (review base)	Flow	Class	HEAD
H1	catalog-api/internal/metrics/health.go:157	`Message: "Database ping failed: " + err.Error()` in a `ComponentHealth` returned by `checkDatabase`, serialised by the unauthenticated `/health/deep` route (registered at `main.go:1107` at base; the comment at `main.go:1052` says "/health and /health/deep stay public")	LEAK	CLOSED (fixed text "Database ping failed", error logged; CAT-PXF23)
H2	catalog-api/internal/monitoring/monitoring.go:251	`"error": err.Error()` in the map built by `HealthChecker.Check`, served by `GinHealthHandler`	LEAK, LATENT (package unwired)	STILL PRESENT, unchanged (see R1)

H1 note: this is the only base instance of the "struct literal" route in `internal/metrics`. Instrument 2 lists exactly two `.Error()` calls in `internal/metrics` and `internal/monitoring` together at the review base (H1, H2), so the route is exhausted for those packages. user_handler.go:171,176 (`msg := err.Error()` echoed in `details`) are the "variable" route and appear in the per-site table as taint rows.

Class 3: helpers that take an error (class-complete list)

Every non-test function in handlers, middleware, internal/handlers, internal/middleware, utils, internal/metrics and internal/monitoring with an error parameter was read (grep of `func ...(err error`). Result at HEAD:

Helper	Does err text reach the client	Verdict
utils.SendErrorResponse	no (log only)	SAFE
handlers.respondInternal / respondBadBody / stdError (client_errors.go)	no (fixed message / status text)	SAFE
handlers.syncFail (client_errors.go:85)	fixed class text in `details` (syncDetails)	SAFE
handlers.validationText / sentinelText / paramRequiredText	authored text only (ValidationError.Msg, sentinel's own text, " parameter required")	VALIDATION
handlers.authClientMessage (auth_handler.go:39)	sentinel text only	VALIDATION
handlers.respondPDFError, writeLogServiceError, writeFixedLogError	fixed generic texts (already so at the review base)	SAFE
handlers.writeJobError (conversion_handler.go:337)	only `ErrConversionJobState` text	VALIDATION
handlers.rootURLMessage / localRootMessage / rootSpecMessage (scan_handler.go:553-578)	authored policy text; the caller's own path is echoed; OS error after "cannot resolve path" is dropped; URL refusal text is generic	VALIDATION (intentional)
handlers.AliasHandler.fail, middleware.useFallback, internal/handlers.smbErrorStatus	no (log only / fixed text)	SAFE
middleware.clientValidationDetails (input_validation.go:184)	only `validationError.public`	VALIDATION
smartrules.jsonErrText	authored ValidationError message built from the caller's own JSON decode error	VALIDATION
services notifyUser (conversion_service.go:958, sync_service.go:1084)	debug log only; `conversionNotifyHook` has no non-test setter	SAFE (not a client sink)

At the review base the helpers that differed: `writeJobError` sent `err.Error()` for `ErrConversionJobState` (same authored text, VALIDATION), `authClientMessage` and the `client_errors.go` helpers did not exist (grep at the review base: 0 hits for `func authClientMessage` and no `client_errors.go`).

Class 4: error text stored in state and served by another route

Found by instrument 3 (flow scan, below) and then traced by reading the writer, the model's JSON tag, the route registration and the gate. All routes are under the `/api/v1` group of `main.go` (authenticated unless a row says otherwise). Line numbers are HEAD (713a4f380) unless marked.

Row	Field and writer	Served by (route, gate)	Scrub	Class
S1	`ScanStatus.Reason`: `fail`/`cancel` in universal_scanner.go:505-522 store `scrubSecrets(err.Error(), secrets)`	`GET /scans`, `GET /scans/:job_id` (`scan_handler.go:539` `"reason": s.Reason`; `RegisterScanRoutes`, storage_routes.go:25-26): any authenticated user. Also `publishScanEvent` (universal_scanner.go:161-194, payload `error`/`reason`) -> `EventBusBridge` (internal/handlers/eventbus_bridge.go:74,96,189-190; wired main.go:681,725) -> a `notification` message passed to `BroadcastToClients`, which routes by `audienceForType` (handlers/websocket_handler.go:74-81, unchanged between 713a4f380 and 9e624f09f): only `media_update` and `asset_update` are `AudienceAuthenticated`, every other type including `notification` is `AudienceAdmin`, and both broadcast paths skip a connection unless `wc.isAdmin` (websocket_handler.go:622 and :653). The WebSocket leg is therefore admin-only; the HTTP leg `GET /scans` is any authenticated user (corrected 2026-11-14, PXF18-FU1; the earlier text said all WebSocket clients)	credentials and URL userinfo only (scan_redact.go); paths, host names and OS error text remain	LEAK (partly scrubbed), open
S2	`GetStorageRootStatus` (scan_handler.go:332,339,345): `resp["error"] = services.ScrubErrorFor(err, root)`, HTTP 200	`/storage-roots/:id/status`, no `RequireAdmin` (storage_routes.go:18): any authenticated user	`ScrubErrorFor` (same scrubber, plus the root's own secrets)	LEAK (partly scrubbed), open
S3	`ListStoragePath` (scan_handler.go:378,384,401): same call	`GET /api/v1/storage/list/*path` (handlers/storage_routes.go:14, `auth.RequireAdmin()`; mounted by `RegisterStorageRoutes(api, ...)`, main.go:1097 at 9e624f09f). There is no `/storage-roots/:id/list` route (grep of `storage-roots/:id/list` in non-test Go: 0 hits); corrected 2026-11-14, PXF18-FU3	same	LEAK to admins (partly scrubbed), accepted by role
S4	`SyncSession.ErrorMessage`: `handleSyncError` (sync_service.go:1053-1066) stores raw `syncError.Error()`; chains such as "failed to get WebDAV client / scan local files / list remote files: %w" (models/user.go:1229, json `error_message,omitempty`)	`GET /sync/sessions`, `GET /sync/sessions/:id`, `GET /sync/history` (main.go:1657,1658,1715): the caller's own sessions, no admin gate	none. Measured: Go's net/http masks the password in a client error (`Get "http://usr:***@127.0.0.1:1/x": dial tcp ...`), so a password in the endpoint URL is masked by the standard client; other userinfo and transport routes are UNCONFIRMED. Host, path and OS text remain	LEAK, open
S5	`ConversionJob.ErrorMessage` (models/user.go:1108, struct `ConversionJob` at :1091, json `error_message,omitempty`; models/file.go:158 is `ScanHistory.ErrorMessage`, a different struct, see "Checked-SAFE surfaces"): conversion_service.go stores `sanitizeConversionError(conversionError.Error())` (reduces paths to base names; line 900 at 9e624f09f)	`GET /conversion/jobs/:id` -> `ConversionHandler.GetJob` -> `ConversionService.authorizedJob` (conversion_service.go:1006): at 9e624f09f the job is returned only to its owner or a system administrator, any other caller gets `ErrConversionJobNotFound`; at 713a4f380 `GetJob` also let a non-owner holding `auth.PermissionViewMedia` read it (conversion_service.go:976-988 at that base; changed by 98e68d37a, CAT-JOB-404-1). `GET /conversion/jobs` -> `ListJobs`: needs `PermissionConversionView` (conversion_handler.go:113) and lists only the caller's own jobs (`WHERE user_id = ?`, repository/conversion_repository.go:278). Routes: main.go:1142-1143 at 9e624f09f. The earlier text "any authenticated user" was wrong (PXF18-FU2)	paths reduced to base names; other text (driver, ffmpeg stderr) remains	LEAK (reduced), open

Row	Field and writer	Served by (route, gate)	Scrub	Class
S6	`WizardStepValidation.Errors[...]` (configuration_service.go:420,430,443): of the four registered validators (database, network, path, email; services/configuration_service.go:45-48) two can return an error, path ("path must be absolute") and email ("invalid email format"), both fixed authored texts, and the database and network validators return nil (lines 945-969; the earlier text said three validators, corrected 2026-11-14, PXF18-FU10); "%s is required" uses the field label	`POST /configuration/wizard/step/:step_id/validate` -> `ConfigurationServiceAdapter.ValidateWizardStep` (handlers/service_adapters.go:76) -> JSON (main.go:1495), gated by `system.configure`	none needed	VALIDATION
S7	`ConfigurationImportResult.ValidationErrors/SkippedSettings` (internal/services/localization_service.go:849,857,877,886,895,904)	no route: `LocalizationHandlers` is constructed only in tests (`NewLocalizationHandlers` has 0 non-test callers; main.go registers no `/wizard/configuration` path)	n/a	LATENT
S8	`BatchResult.Errors`, `SyncResult.Errors` (services/webdav_client.go:340,361,429,450,482,502)	no non-test caller of `UploadBatch`, `DownloadBatch`, `SyncDirectory` outside the file (grep: only webdav_client_test.go)	n/a	LATENT
S9	`retry` result `Error` (internal/recovery/retry.go:401)	`internal/recovery` has no non-test importer (grep `internal/recovery"`: 0 hits; "digital.vasic.recovery" in internal/modules/registry.go is a name string)	n/a	LATENT
S10	`liveevidence.Record.ResolveError` (internal/liveevidence/liveevidence.go:158)	written to evidence files by Go tests; no non-test importer	n/a	SAFE (not a client sink)
S11	`log_management_service.logError` (log_management_service.go:170,194,204): `message` is formatted from `%v` of errors	the function only calls the structured logger (lines 724-732)	n/a	SAFE
S12	`notifyUser` (conversion_service.go, sync_service.go), `logSyncError`/`logSkippedFile`	log or debug output only	n/a	SAFE

Instrument 3 reports 362 flow rows at HEAD (402 at the review base; `challenges/` is skipped, see R3). Both counts are AFTER `sort -u`: the raw output has 365 rows at HEAD and 405 at the review base, because the scan emits 3 duplicate rows at each base (main.go `callarg:metricsSkipIfScrape`, `callarg:router.GET`, `callarg:pprofGroup.Use`, all `T:jwtMiddleware`; main.go:1060 and :1076 at HEAD, :1053 and :1069 at the review base; PXF18-FU5, re-run 2026-11-14). The field, key-value and call-argument rows of the files named in S1 to S12 were each traced above. The remaining rows (mostly constant or authored text assigned to fields, and calls into logging helpers) were NOT each traced one by one: UNCONFIRMED whether any of them reaches a client through a route this document did not look at (see Limits).

HEAD residual (everything still in the class at 713a4f380)

Id	Site	Class	Note
R1	catalog-api/internal/monitoring/monitoring.go:251 `"error": err.Error()` (GinHealthHandler)	LATENT-OPEN	the package has no non-test importer; the leak becomes real the moment it is wired. Decide: sanitise or delete (delete needs operator confirmation, §11.4.122 / §11.4.124)
R2	OCU-CUDA-Sidecar/internal/server/server.go:110,115,128,133,150,154,159,172,176,180,185,198,203,216,221 (15 sites, `http.Error(w, err.Error()...)` and `"bad request: "+err.Error()`)	LEAK	a separate Go module, same 15 sites at both bases, outside the catalog-api scan the old doc used. No in-module caller of `ListenAndServe` and no auth in `server.go`; whether the process is reachable by untrusted clients is UNCONFIRMED
R3	catalog-api/handlers/challenge.go and the `challenges/` package: 155 `.Error()` calls (74 `CreateResult`, 20 `image_quality_final.go`, 16 `image_quality.go`, 13 `ch131_middleware_challenges.go`, ...) feed challenge result records that `ChallengeHandler.GetResults`, `RunChallenge` and `RunAll` return as JSON (`handlers/challenge.go:115,185`); `POST` run routes are admin-gated (`main.go:1670-1672`), the gating of `GET /api/v1/challenges/results` was not established	UNCONFIRMED	not classified per site; decide whether result text is operator-only

Id	Site	Class	Note
R4	VALIDATION sentinel/authored sites: playlist_handler.go:95,99,188,273,320; collection_handler.go:335,351 (`validationText(err)` as the SendErrorResponse message); service_handlers.go:199,258,298 (`paramRequiredText(err)`); conversion_handler.go:72-73 (row B62: `reason` is the text after `ErrInvalidConversionRequest`, produced by `conversionRequestProblem`, a short client-safe reason, conversion_service.go:82-83 and :1448; line 73 at 713a4f380), conversion_handler.go:344; scan_handler.go:178,186,194,448; middleware/input_validation.go:403 (`clientValidationDetails(err)`)	VALIDATION	authored text; scan rows echo the caller's own path; these rows are NOT closed, the text still reaches the client
R5	`handlers/scan_handler.go` localRootMessage keeps `msg[:i] + ": cannot resolve path"`	VALIDATION	reads: the prefix of an authored policy sentence plus the path the caller sent
R6	Class 4 rows S1, S2, S4, S5: `ScanStatus.Reason` (HTTP `GET /scans`: any authenticated user; the WebSocket broadcast is admin-only, see S1), `GetStorageRootStatus` `error`, `SyncSession.ErrorMessage`, `ConversionJob.ErrorMessage`	LEAK (partly scrubbed)	open; S1 (HTTP), S2 and S4 are served to authenticated users, S5 to the job's owner or an administrator (see S5), and S1's WebSocket leg only to administrators. Decision owed per field: fixed class text for non-admin routes and the full text only for admins or the log. Not changed by this document

Everything else the sink scan reports at HEAD is SAFE by the helper table (111 `SendErrorResponse` + 16 `respondInternal/respondBadBody` + 5 `respondPDFError` + 5 `writeLogServiceError` + 5 `writeJobError` + 2 `c.Error` bare-err rows; the `c.Error` rows are ErrorTypePrivate, which ErrorHandler answers with a fixed message at HEAD: `internal/middleware/middleware.go:97-100`).

Census (instrument 2, all non-test code under catalog-api)

Group	Review base	HEAD
production code	175	97
challenges/ (harness, see R3)	155	155
internal/tests (test helpers)	3	3
total	333	255

HEAD production-code contexts (97): branching on text (`strings.Contains` 42, `strings.ToLower` 7, `strings.TrimPrefix` 2, `branch` 6) = 57, read at the branch: the compared text does not itself go to a response (stored-then-served flows are in Class 4); assignment 14, return 9, key-value 6, scrub/log helpers (`scrubSecrets` 5, `logSkippedFile` 2, `sanitizeConversionError` 1, `panic` 1, `paramRequiredRE.FindString` 1), `fmt.Sprintf` 1 (sync_service.go:1066, notifyUser, SAFE). `scrubSecrets` at universal_scanner.go:510,520 stores the scrubbed text in `ScanStatus.Reason`, which IS served (row S1); sync_service.go:1057-1058 `handleSyncError` stores the raw text in `SyncSession.ErrorMessage`, which IS served (row S4). The earlier sentences that none of these reach a response are withdrawn. The 14 assign + 9 return + 6 kv rows were read with the helper table above; the two that carry text to a body are the sentinel rows in R4 and the unwired R1.

Review-base totals (instrument 1, 226 rows after removing `strings.*` callee noise) plus H1/H2

Class	SendErrorResponse family	other sinks
LEAK	109	49
VALIDATION	1	33
LATENT	0	15
SAFE	0	19
total	110	116

HEAD status of the 226 rows: CLOSED-BY-HELPER 109; CLOSED (err text no longer sent) 64 (49 LEAK + 15 LATENT); VALIDATION rows where `err.Error()` is no longer sent but authored text still is (NOT CLOSED) 32; unchanged (SAFE 19, VALIDATION 2) 21. This count covers the direct-sink rows only. It is NOT a claim that no leak remains: Class 4 rows S1 to S5 are still served at HEAD with credential-only or path-only scrubbing, and R1 to R3 are open.

Per-site table (review base 9ed6577b, with HEAD status)

Row ids B1.. are sorted by file and line. `st` is the status argument when the call has one.

#	File:line	Function	Sink	st	Expression	Class	Reason	HEAD
B1	catalog-api/handlers/asset_handler.go:91	RequestAsset	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	binding/JSON-decode error; names Go struct fields/types, echoes malformed input	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B2	catalog-api/handlers/auth_handler.go:36	LoginGin	c.JSON	http.StatusForbidden	`E:err.Error()`	VALIDATION	branch runs only when errors.Is(err, ErrAccountPending); Login returns the bare sentinel (auth_service.go:152 at base), so the body is its authored text "account pending approval"	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B3	catalog-api/handlers/auth_handler.go:39	LoginGin	c.JSON	http.StatusUnauthorized	`E:err.Error()`	LEAK	401 branch: Login returns sentinels (invalid credentials, locked, disabled; authored) and also "failed to get user: %w" wrapping repository text (auth_service.go:132 at base); the repository text reaches the body	CLOSED (err text no longer sent)
B4	catalog-api/handlers/auth_handler.go:59	RefreshTokenGin	c.JSON	http.StatusUnauthorized	`E:err.Error()`	LEAK	(unresolved in review base) auth failure text; wording decides user-enumeration exposure	CLOSED (err text no longer sent)
B5	catalog-api/handlers/auth_handler.go:76	LogoutGin	c.JSON	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B6	catalog-api/handlers/auth_handler.go:203	RegisterGin	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	binding/JSON-decode error; names Go struct fields/types, echoes malformed input	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B7	catalog-api/handlers/auth_handler.go:321	Login	http.Error	http.StatusUnauthorized	`E:err.Error()`	LATENT	(unresolved in review base) auth failure text; wording decides user-enumeration exposure. UNROUTED: net/http variant; main.go:1210-1220 registers only the Gin variants (LoginGin, RefreshTokenGin, LogoutGin, ...)	CLOSED (err text no longer sent)
B8	catalog-api/handlers/auth_handler.go:346	RefreshToken	http.Error	http.StatusUnauthorized	`E:err.Error()`	LATENT	(unresolved in review base) auth failure text; wording decides user-enumeration exposure. UNROUTED: net/http variant; main.go:1210-1220 registers only the Gin variants (LoginGin, RefreshTokenGin, LogoutGin, ...)	CLOSED (err text no longer sent)
B9	catalog-api/handlers/auth_handler.go:368	Logout	http.Error	http.StatusInternalServerError	`E:err.Error()`	LATENT	service/repo/DB error text sent verbatim on 5xx or generic status. UNROUTED: net/http variant; main.go:1210-1220 registers only the Gin variants (LoginGin, RefreshTokenGin, LogoutGin, ...)	CLOSED (err text no longer sent)

#	File:line	Function	Sink	st	Expression	Class	Reason	HEAD
B10	catalog-api/handlers/auth_handler.go:390	LogoutAll	http.Error	http.StatusInternalServerError	`E:err.Error()`	LATENT	service/repo/DB error text sent verbatim on 5xx or generic status. UNROUTED: net/http variant; main.go:1210-1220 registers only the Gin variants (LoginGin, RefreshTokenGin, LogoutGin, ...)	CLOSED (err text no longer sent)
B11	catalog-api/handlers/auth_handler.go:440	ChangePassword	http.Error	http.StatusBadRequest	`E:err.Error()`	VALIDATION	deliberate validation text (may echo user input). The net/http `AuthHandler.ChangePassword` is also unrouted (main.go:1222 uses `mediaQueryHandler.ChangePassword`); kept VALIDATION (authored text), not recounted as LATENT	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B12	catalog-api/handlers/auth_handler.go:446	ChangePassword	http.Error	http.StatusBadRequest	`E:err.Error()`	LATENT	ChangePassword returns wrapped "failed to get user/update password: %w" chains (auth_service.go:326,348,372) on 400. UNROUTED: net/http variant; main.go:1210-1220 registers only the Gin variants (LoginGin, RefreshTokenGin, LogoutGin, ...)	CLOSED (err text no longer sent)
B13	catalog-api/handlers/auth_handler.go:468	GetActiveSessions	http.Error	http.StatusInternalServerError	`E:err.Error()`	LATENT	service/repo/DB error text sent verbatim on 5xx or generic status. UNROUTED: net/http variant; main.go:1210-1220 registers only the Gin variants (LoginGin, RefreshTokenGin, LogoutGin, ...)	CLOSED (err text no longer sent)
B14	catalog-api/handlers/auth_handler.go:507	DeactivateSession	http.Error	http.StatusInternalServerError	`E:err.Error()`	LATENT	service/repo/DB error text sent verbatim on 5xx or generic status. UNROUTED: net/http variant; main.go:1210-1220 registers only the Gin variants (LoginGin, RefreshTokenGin, LogoutGin, ...)	CLOSED (err text no longer sent)
B15	catalog-api/handlers/browse.go:41	GetStorageRoots	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B16	catalog-api/handlers/browse.go:120	BrowseDirectory	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B17	catalog-api/handlers/browse.go:164	GetFileInfo	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B18	catalog-api/handlers/browse.go:217	GetDirectorySizes	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B19	catalog-api/handlers/browse.go:270	GetDirectoryDuplicates	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B20	catalog-api/handlers/challenge.go:103	RunChallenge	utils.SendErrorResponse	http.StatusNotFound	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B21	catalog-api/handlers/challenge.go:106	RunChallenge	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B22	catalog-api/handlers/challenge.go:166	RunAll	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B23	catalog-api/handlers/challenge.go:234	RunByCategory	utils.SendErrorResponse	http.StatusNotFound	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)

#	File:line	Function	Sink	st	Expression	Class	Reason	HEAD
B24	catalog-api/handlers/challenge.go:237	RunByCategory	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B25	catalog-api/handlers/collection_handler.go:100	loadAccessible	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B26	catalog-api/handlers/collection_handler.go:105	loadAccessible	utils.SendErrorResponse	http.StatusNotFound	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B27	catalog-api/handlers/collection_handler.go:160	ListCollections	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B28	catalog-api/handlers/collection_handler.go:269	GetCollection	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B29	catalog-api/handlers/collection_handler.go:315	ListCollectionItems	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B30	catalog-api/handlers/collection_handler.go:335	parseSmartRules	utils.SendErrorResponse	http.StatusBadRequest	`E:err.Error()`	VALIDATION	parseSmartRules: smartrules.ValidationError text passed as message (authored); non-validation errors take the 500 row at :337	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B31	catalog-api/handlers/collection_handler.go:337	parseSmartRules	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B32	catalog-api/handlers/collection_handler.go:353	verifyRules	utils.SendErrorResponse	-	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B33	catalog-api/handlers/collection_handler.go:374	CreateCollection	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B34	catalog-api/handlers/collection_handler.go:429	CreateCollection	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B35	catalog-api/handlers/collection_handler.go:452	CreateCollection	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B36	catalog-api/handlers/collection_handler.go:476	UpdateCollection	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B37	catalog-api/handlers/collection_handler.go:480	UpdateCollection	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B38	catalog-api/handlers/collection_handler.go:539	UpdateCollection	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B39	catalog-api/handlers/collection_handler.go:561	DeleteCollection	utils.SendErrorResponse	http.StatusNotFound	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B40	catalog-api/handlers/collection_handler.go:564	DeleteCollection	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)

#	File:line	Function	Sink	st	Expression	Class	Reason	HEAD
B41	catalog-api/handlers/comic_pages_handler.go:106	resolveArchive	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B42	catalog-api/handlers/comic_pages_handler.go:113	resolveArchive	utils.SendErrorResponse	http.StatusNotFound	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B43	catalog-api/handlers/comic_pages_handler.go:116	resolveArchive	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B44	catalog-api/handlers/comic_pages_handler.go:434	ListComicPages	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B45	catalog-api/handlers/comic_pages_handler.go:467	GetComicPage	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B46	catalog-api/handlers/comic_pages_handler.go:478	GetComicPage	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B47	catalog-api/handlers/comic_pages_handler.go:492	GetComicPage	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B48	catalog-api/handlers/comic_pages_handler.go:515	listCBRPages	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B49	catalog-api/handlers/comic_pages_handler.go:522	listCBRPages	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B50	catalog-api/handlers/comic_pages_handler.go:552	getCBRPage	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B51	catalog-api/handlers/comic_pages_handler.go:559	getCBRPage	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B52	catalog-api/handlers/comic_pages_handler.go:570	getCBRPage	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B53	catalog-api/handlers/comic_pages_handler.go:582	getCBRPage	utils.SendErrorResponse	http.StatusInternalServerError	`nerr`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B54	catalog-api/handlers/comic_pages_handler.go:592	getCBRPage	utils.SendErrorResponse	http.StatusInternalServerError	`rerr`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B55	catalog-api/handlers/configuration_handler.go:57	GetWizardStep	http.Error	http.StatusNotFound	`E:err.Error()`	VALIDATION	"wizard step not found: " authored, echoes the caller step id	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B56	catalog-api/handlers/configuration_handler.go:77	ValidateWizardStep	http.Error	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B57	catalog-api/handlers/configuration_handler.go:102	SaveWizardProgress	http.Error	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)

#	File:line	Function	Sink	st	Expression	Class	Reason	HEAD
B58	catalog-api/handlers/configuration_handler.go:121	GetWizardProgress	http.Error	http.StatusNotFound	`E:err.Error()`	LEAK	GetWizardProgress returns repo text unchanged (driver / sql: no rows) on 404	CLOSED (err text no longer sent)
B59	catalog-api/handlers/configuration_handler.go:145	CompleteWizard	http.Error	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B60	catalog-api/handlers/configuration_handler.go:173	GetConfiguration	http.Error	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B61	catalog-api/handlers/configuration_handler.go:203	TestConfiguration	http.Error	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B62	catalog-api/handlers/conversion_handler.go:73	CreateJob	c.JSON	http.StatusBadRequest	`T:reason<-strings.TrimPrefix(strings.TrimPrefix(err.Error(), models.ErrInvalidConversionRe`	VALIDATION	INDIRECT: reason is the text after ErrInvalidConversionRequest (authored)	unchanged; class unchanged (VALIDATION)
B63	catalog-api/handlers/conversion_handler.go:99	GetJob	writeJobError	-	`err`	SAFE	helper answers fixed texts; only the ErrConversionJobState text is sent (row :344)	unchanged; class unchanged (SAFE)
B64	catalog-api/handlers/conversion_handler.go:185	CancelJob	writeJobError	-	`err`	SAFE	helper answers fixed texts; only the ErrConversionJobState text is sent (row :344)	unchanged; class unchanged (SAFE)
B65	catalog-api/handlers/conversion_handler.go:246	DeleteJob	writeJobError	-	`err`	SAFE	helper answers fixed texts; only the ErrConversionJobState text is sent (row :344)	unchanged; class unchanged (SAFE)
B66	catalog-api/handlers/conversion_handler.go:282	RetryJob	writeJobError	-	`err`	SAFE	helper answers fixed texts; only the ErrConversionJobState text is sent (row :344)	unchanged; class unchanged (SAFE)
B67	catalog-api/handlers/conversion_handler.go:307	DownloadJobFile	writeJobError	-	`err`	SAFE	helper answers fixed texts; only the ErrConversionJobState text is sent (row :344)	unchanged; class unchanged (SAFE)
B68	catalog-api/handlers/conversion_handler.go:344	writeJobError	c.JSON	http.StatusConflict	`E:err.Error()`	VALIDATION	ErrConversionJobState sentinel text, authored	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B69	catalog-api/handlers/error_reporting_handler.go:85	ReportError	http.Error	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B70	catalog-api/handlers/error_reporting_handler.go:123	ReportCrash	http.Error	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B71	catalog-api/handlers/error_reporting_handler.go:152	GetErrorReport	http.Error	http.StatusNotFound	`E:err.Error()`	LEAK	"failed to get error report: %w" wraps the repository error (driver text)	CLOSED (err text no longer sent)
B72	catalog-api/handlers/error_reporting_handler.go:181	GetCrashReport	http.Error	http.StatusNotFound	`E:err.Error()`	LEAK	"failed to get crash report: %w" wraps the repository error (driver text)	CLOSED (err text no longer sent)
B73	catalog-api/handlers/error_reporting_handler.go:205	ListErrorReports	http.Error	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B74	catalog-api/handlers/error_reporting_handler.go:232	ListCrashReports	http.Error	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)

#	File:line	Function	Sink	st	Expression	Class	Reason	HEAD
B75	catalog-api/handlers/error_reporting_handler.go:273	UpdateErrorStatus	http.Error	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B76	catalog-api/handlers/error_reporting_handler.go:310	UpdateCrashStatus	http.Error	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B77	catalog-api/handlers/error_reporting_handler.go:332	GetErrorStatistics	http.Error	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B78	catalog-api/handlers/error_reporting_handler.go:355	GetCrashStatistics	http.Error	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B79	catalog-api/handlers/error_reporting_handler.go:378	GetSystemHealth	http.Error	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B80	catalog-api/handlers/error_reporting_handler.go:407	UpdateConfiguration	http.Error	http.StatusInternalServerError	`E:err.Error()`	LATENT	service/repo/DB error text sent verbatim on 5xx or generic status. UNROUTED: main.go registers none of UpdateConfiguration/GetConfiguration/CleanupOldReports/ExportReports of ErrorReportingHandler (grep main.go: 0 hits)	CLOSED (err text no longer sent)
B81	catalog-api/handlers/error_reporting_handler.go:432	GetConfiguration	http.Error	http.StatusInternalServerError	`E:err.Error()`	LATENT	service/repo/DB error text sent verbatim on 5xx or generic status. UNROUTED: main.go registers none of UpdateConfiguration/GetConfiguration/CleanupOldReports/ExportReports of ErrorReportingHandler (grep main.go: 0 hits)	CLOSED (err text no longer sent)
B82	catalog-api/handlers/error_reporting_handler.go:469	CleanupOldReports	http.Error	http.StatusInternalServerError	`E:err.Error()`	LATENT	service/repo/DB error text sent verbatim on 5xx or generic status. UNROUTED: main.go registers none of UpdateConfiguration/GetConfiguration/CleanupOldReports/ExportReports of ErrorReportingHandler (grep main.go: 0 hits)	CLOSED (err text no longer sent)
B83	catalog-api/handlers/error_reporting_handler.go:495	ExportReports	http.Error	http.StatusInternalServerError	`E:err.Error()`	LATENT	service/repo/DB error text sent verbatim on 5xx or generic status. UNROUTED: main.go registers none of UpdateConfiguration/GetConfiguration/CleanupOldReports/ExportReports of ErrorReportingHandler (grep main.go: 0 hits)	CLOSED (err text no longer sent)
B84	catalog-api/handlers/log_management_handler.go:222	GetLogEntries	writeLogServiceError	-	`err`	SAFE	helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised	unchanged; class unchanged (SAFE)
B85	catalog-api/handlers/log_management_handler.go:254	CreateLogShare	writeLogServiceError	-	`err`	SAFE	helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised	unchanged; class unchanged (SAFE)
B86	catalog-api/handlers/log_management_handler.go:322	RevokeLogShare	writeLogServiceError	-	`err`	SAFE	helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised	unchanged; class unchanged (SAFE)

#	File:line	Function	Sink	st	Expression	Class	Reason	HEAD
B87	catalog-api/handlers/log_management_handler.go:355	ExportLogs	writeLogServiceError	-	`err`	SAFE	helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised	unchanged; class unchanged (SAFE)
B88	catalog-api/handlers/log_management_handler.go:475	AnalyzeLogs	writeLogServiceError	-	`err`	SAFE	helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised	unchanged; class unchanged (SAFE)
B89	catalog-api/handlers/media_browse_handler.go:83	SearchMedia	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B90	catalog-api/handlers/media_browse_handler.go:106	GetMediaStats	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B91	catalog-api/handlers/media_browse_handler.go:127	GetMediaStats	c.Error	-	`rowsErr`	SAFE	c.Error(rowsErr) is ErrorTypePrivate: ErrorHandler default branch answers a fixed message	unchanged; class unchanged (SAFE)
B92	catalog-api/handlers/media_browse_handler.go:158	GetMediaStats	c.Error	-	`rowsErr`	SAFE	same as :127	unchanged; class unchanged (SAFE)
B93	catalog-api/handlers/media_entity_handler.go:77	SearchEntities	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B94	catalog-api/handlers/media_entity_handler.go:85	SearchEntities	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B95	catalog-api/handlers/media_entity_handler.go:227	ListEntities	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B96	catalog-api/handlers/media_entity_handler.go:239	ListEntities	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B97	catalog-api/handlers/media_entity_handler.go:260	GetEntity	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B98	catalog-api/handlers/media_entity_handler.go:266	GetEntity	utils.SendErrorResponse	http.StatusNotFound	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B99	catalog-api/handlers/media_entity_handler.go:306	GetEntityChildren	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B100	catalog-api/handlers/media_entity_handler.go:315	GetEntityChildren	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B101	catalog-api/handlers/media_entity_handler.go:336	GetEntityFiles	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B102	catalog-api/handlers/media_entity_handler.go:342	GetEntityFiles	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B103	catalog-api/handlers/media_entity_handler.go:358	GetEntityMetadata	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)

#	File:line	Function	Sink	st	Expression	Class	Reason	HEAD
B104	catalog-api/handlers/media_entity_handler.go:364	GetEntityMetadata	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B105	catalog-api/handlers/media_entity_handler.go:379	GetEntityDuplicates	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B106	catalog-api/handlers/media_entity_handler.go:385	GetEntityDuplicates	utils.SendErrorResponse	http.StatusNotFound	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B107	catalog-api/handlers/media_entity_handler.go:391	GetEntityDuplicates	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B108	catalog-api/handlers/media_entity_handler.go:415	GetEntityTypes	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B109	catalog-api/handlers/media_entity_handler.go:457	BrowseByType	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B110	catalog-api/handlers/media_entity_handler.go:463	BrowseByType	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B111	catalog-api/handlers/media_entity_handler.go:496	RefreshEntityMetadata	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B112	catalog-api/handlers/media_entity_handler.go:1158	entityMustExist	utils.SendErrorResponse	http.StatusNotFound	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B113	catalog-api/handlers/media_entity_handler.go:1160	entityMustExist	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B114	catalog-api/handlers/media_entity_handler.go:1173	UpdateUserMetadata	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B115	catalog-api/handlers/media_entity_handler.go:1186	UpdateUserMetadata	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B116	catalog-api/handlers/media_entity_handler.go:1217	UpdateUserMetadata	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B117	catalog-api/handlers/media_entity_handler.go:1246	StreamEntity	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B118	catalog-api/handlers/media_entity_handler.go:1260	StreamEntity	utils.SendErrorResponse	http.StatusNotFound	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B119	catalog-api/handlers/media_entity_handler.go:1263	StreamEntity	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B120	catalog-api/handlers/media_entity_handler.go:1283	DownloadEntity	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B121	catalog-api/handlers/media_entity_handler.go:1291	DownloadEntity	utils.SendErrorResponse	http.StatusNotFound	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)

#	File:line	Function	Sink	st	Expression	Class	Reason	HEAD
B122	catalog-api/handlers/media_entity_handler.go:1320	DownloadEntity	utils.SendErrorResponse	http.StatusNotFound	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B123	catalog-api/handlers/media_entity_handler.go:1323	DownloadEntity	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B124	catalog-api/handlers/media_entity_handler.go:1343	GetInstallInfo	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B125	catalog-api/handlers/media_entity_handler.go:1349	GetInstallInfo	utils.SendErrorResponse	http.StatusNotFound	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B126	catalog-api/handlers/media_entity_handler.go:1369	GetInstallInfo	utils.SendErrorResponse	http.StatusNotFound	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B127	catalog-api/handlers/media_entity_handler.go:1402	ListDuplicateGroups	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B128	catalog-api/handlers/media_handler.go:46	GetMediaByID	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B129	catalog-api/handlers/media_handler.go:87	GetMediaByID	utils.SendErrorResponse	http.StatusNotFound	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B130	catalog-api/handlers/media_handler.go:89	GetMediaByID	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B131	catalog-api/handlers/media_handler.go:148	UpdateWatchProgress	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B132	catalog-api/handlers/media_handler.go:155	UpdateWatchProgress	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B133	catalog-api/handlers/media_handler.go:205	UpdateWatchProgress	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B134	catalog-api/handlers/media_handler.go:238	UpdateFavoriteStatus	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B135	catalog-api/handlers/media_handler.go:245	UpdateFavoriteStatus	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B136	catalog-api/handlers/media_handler.go:267	UpdateFavoriteStatus	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B137	catalog-api/handlers/media_handler.go:274	UpdateFavoriteStatus	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B138	catalog-api/handlers/pdf_pages_handler.go:101	resolvePDF	respondPDFError	-	`err`	SAFE	helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised	unchanged; class unchanged (SAFE)

#	File:line	Function	Sink	st	Expression	Class	Reason	HEAD
B139	catalog-api/handlers/pdf_pages_handler.go:282	ListPdfPages	respondPDFError	-	`err`	SAFE	helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised	unchanged; class unchanged (SAFE)
B140	catalog-api/handlers/pdf_pages_handler.go:289	ListPdfPages	respondPDFError	-	`err`	SAFE	helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised	unchanged; class unchanged (SAFE)
B141	catalog-api/handlers/pdf_pages_handler.go:322	GetPdfPage	respondPDFError	-	`err`	SAFE	helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised	unchanged; class unchanged (SAFE)
B142	catalog-api/handlers/pdf_pages_handler.go:329	GetPdfPage	respondPDFError	-	`err`	SAFE	helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised	unchanged; class unchanged (SAFE)
B143	catalog-api/handlers/playback_handler.go:79	StartSession	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	binding/JSON-decode error; names Go struct fields/types, echoes malformed input	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B144	catalog-api/handlers/playback_handler.go:96	StartSession	c.JSON	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B145	catalog-api/handlers/playback_handler.go:106	ProgressSession	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	binding/JSON-decode error; names Go struct fields/types, echoes malformed input	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B146	catalog-api/handlers/playback_handler.go:118	ProgressSession	c.JSON	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B147	catalog-api/handlers/playback_handler.go:128	EndSession	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	binding/JSON-decode error; names Go struct fields/types, echoes malformed input	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B148	catalog-api/handlers/playback_handler.go:143	EndSession	c.JSON	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B149	catalog-api/handlers/playback_handler.go:170	GetProgressForEntity	c.JSON	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B150	catalog-api/handlers/playback_handler.go:193	ListHistoryForEntity	c.JSON	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)

#	File:line	Function	Sink	st	Expression	Class	Reason	HEAD
B151	catalog-api/handlers/playlist_handler.go:95	CreatePlaylist	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	smartrules.IsValidation branch: authored ValidationError text (may echo caller JSON)	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B152	catalog-api/handlers/playlist_handler.go:99	CreatePlaylist	c.JSON	http.StatusConflict	`E:err.Error()`	VALIDATION	ErrSmartLimit sentinel with authored count text	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B153	catalog-api/handlers/playlist_handler.go:188	UpdatePlaylist	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	smartrules.IsValidation branch: authored text	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B154	catalog-api/handlers/playlist_handler.go:273	AddItem	c.JSON	http.StatusConflict	`E:err.Error()`	VALIDATION	ErrSmartPlaylistItems sentinel, authored	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B155	catalog-api/handlers/playlist_handler.go:320	RemoveItem	c.JSON	http.StatusConflict	`E:err.Error()`	VALIDATION	ErrSmartPlaylistItems sentinel, authored	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B156	catalog-api/handlers/recommendation_handler.go:54	GetSimilarItems	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B157	catalog-api/handlers/recommendation_handler.go:76	GetSimilarItems	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B158	catalog-api/handlers/recommendation_handler.go:93	GetSimilarItems	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B159	catalog-api/handlers/recommendation_handler.go:115	GetTrendingItems	utils.SendErrorResponse	http.StatusNotImplemented	`ErrRecommendationsNotImplemented`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B160	catalog-api/handlers/recommendation_handler.go:135	GetPersonalizedRecommendations	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B161	catalog-api/handlers/recommendation_handler.go:152	GetPersonalizedRecommendations	utils.SendErrorResponse	http.StatusNotImplemented	`ErrRecommendationsNotImplemented`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B162	catalog-api/handlers/scan_handler.go:153	CreateStorageRoot	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	binding/JSON-decode error; names Go struct fields/types, echoes malformed input	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)

#	File:line	Function	Sink	st	Expression	Class	Reason	HEAD
B163	catalog-api/handlers/scan_handler.go:177	CreateStorageRoot	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	deliberate validation text (may echo user input)	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B164	catalog-api/handlers/scan_handler.go:185	CreateStorageRoot	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	deliberate validation text (may echo user input)	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B165	catalog-api/handlers/scan_handler.go:193	CreateStorageRoot	c.JSON	http.StatusForbidden	`E:err.Error()`	LEAK	ErrLocalRootNotAllowed text includes "cannot resolve path: " and server-side policy detail (local_policy.go:54,48,63)	CLOSED (err text no longer sent)
B166	catalog-api/handlers/scan_handler.go:227	CreateStorageRoot	c.JSON	http.StatusInternalServerError	`F:fmt.Sprintf("failed to create storage root: %v", insertErr)`	LEAK	%v of DB/scanner error wrapped into body (SQL/driver detail)	CLOSED (err text no longer sent)
B167	catalog-api/handlers/scan_handler.go:247	GetStorageRoots	c.JSON	http.StatusInternalServerError	`F:fmt.Sprintf("failed to query storage roots: %v", err)`	LEAK	%v of DB/scanner error wrapped into body (SQL/driver detail)	CLOSED (err text no longer sent)
B168	catalog-api/handlers/scan_handler.go:427	QueueScan	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	binding/JSON-decode error; names Go struct fields/types, echoes malformed input	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B169	catalog-api/handlers/scan_handler.go:441	QueueScan	c.JSON	http.StatusNotFound	`F:fmt.Sprintf("storage root not found: %v", err)`	LEAK	%v of DB/scanner error wrapped into body (SQL/driver detail)	CLOSED (err text no longer sent)
B170	catalog-api/handlers/scan_handler.go:447	QueueScan	c.JSON	http.StatusForbidden	`E:err.Error()`	LEAK	same producer as :193 (CheckLocalRoot)	CLOSED (err text no longer sent)
B171	catalog-api/handlers/scan_handler.go:463	QueueScan	c.JSON	http.StatusServiceUnavailable	`F:fmt.Sprintf("failed to queue scan: %v", err)`	LEAK	%v of DB/scanner error wrapped into body (SQL/driver detail)	CLOSED (err text no longer sent)
B172	catalog-api/handlers/search.go:100	SearchFiles	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B173	catalog-api/handlers/search.go:108	SearchFiles	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B174	catalog-api/handlers/search.go:155	SearchFiles	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B175	catalog-api/handlers/search.go:258	SearchDuplicates	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B176	catalog-api/handlers/search.go:287	AdvancedSearch	utils.SendErrorResponse	http.StatusBadRequest	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B177	catalog-api/handlers/search.go:328	AdvancedSearch	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B178	catalog-api/handlers/service_handlers.go:199	CreateReport	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	ErrUnsupportedReportType sentinel, authored	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)

#	File:line	Function	Sink	st	Expression	Class	Reason	HEAD
B179	catalog-api/handlers/service_handlers.go:258	GetUsageReport	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	body only reached when text matches "parameter required" (deliberate); other errors get generic 500	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B180	catalog-api/handlers/service_handlers.go:298	GetPerformanceReport	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	body only reached when text matches "parameter required" (deliberate); other errors get generic 500	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B181	catalog-api/handlers/stats.go:42	GetOverallStats	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B182	catalog-api/handlers/stats.go:82	GetSmbRootStats	utils.SendErrorResponse	http.StatusNotFound	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B183	catalog-api/handlers/stats.go:85	GetSmbRootStats	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B184	catalog-api/handlers/stats.go:119	GetFileTypeStats	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B185	catalog-api/handlers/stats.go:146	GetSizeDistribution	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B186	catalog-api/handlers/stats.go:173	GetDuplicateStats	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B187	catalog-api/handlers/stats.go:214	GetTopDuplicateGroups	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B188	catalog-api/handlers/stats.go:248	GetAccessPatterns	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B189	catalog-api/handlers/stats.go:282	GetGrowthTrends	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B190	catalog-api/handlers/stats.go:321	GetScanHistory	utils.SendErrorResponse	http.StatusInternalServerError	`err`	LEAK	err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status	CLOSED-BY-HELPER (details no longer set; err logged)
B191	catalog-api/handlers/sync_handler.go:40	CreateEndpoint	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	binding/JSON-decode error; names Go struct fields/types, echoes malformed input	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B192	catalog-api/handlers/sync_handler.go:52	CreateEndpoint	c.JSON	-	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B193	catalog-api/handlers/sync_handler.go:69	GetUserEndpoints	c.JSON	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B194	catalog-api/handlers/sync_handler.go:103	GetEndpoint	c.JSON	-	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)

#	File:line	Function	Sink	st	Expression	Class	Reason	HEAD
B195	catalog-api/handlers/sync_handler.go:126	UpdateEndpoint	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	binding/JSON-decode error; names Go struct fields/types, echoes malformed input	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B196	catalog-api/handlers/sync_handler.go:142	UpdateEndpoint	c.JSON	-	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B197	catalog-api/handlers/sync_handler.go:171	DeleteEndpoint	c.JSON	-	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B198	catalog-api/handlers/sync_handler.go:202	StartSync	c.JSON	-	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B199	catalog-api/handlers/sync_handler.go:234	GetUserSessions	c.JSON	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B200	catalog-api/handlers/sync_handler.go:268	GetSession	c.JSON	-	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B201	catalog-api/handlers/sync_handler.go:288	ScheduleSync	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	binding/JSON-decode error; names Go struct fields/types, echoes malformed input	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B202	catalog-api/handlers/sync_handler.go:304	ScheduleSync	c.JSON	-	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B203	catalog-api/handlers/sync_handler.go:341	GetSyncStatistics	c.JSON	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B204	catalog-api/handlers/sync_handler.go:371	CleanupOldSessions	c.JSON	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B205	catalog-api/handlers/user_handler.go:121	CreateUser	http.Error	http.StatusBadRequest	`E:err.Error()`	VALIDATION	deliberate validation text (may echo user input)	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B206	catalog-api/handlers/user_handler.go:171	CreateUser	http.Error	http.StatusConflict	`T:msg<-err.Error()`	LEAK	INDIRECT: msg := err.Error() is sent as "details" on 409 and 500 (user_handler.go:172,177 at base): UNIQUE/PG duplicate-key text and any repo error	CLOSED (details now fixed text; msg only branches, user_handler.go:168-179)
B207	catalog-api/handlers/user_handler.go:176	CreateUser	http.Error	http.StatusInternalServerError	`T:msg<-err.Error()`	LEAK	INDIRECT: same msg as :171 sent as details on 500	CLOSED (details now fixed text; msg only branches, user_handler.go:168-179)
B208	catalog-api/handlers/user_handler.go:624	ResetPassword	http.Error	http.StatusBadRequest	`E:err.Error()`	VALIDATION	deliberate validation text (may echo user input)	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B209	catalog-api/handlers/user_handler.go:630	ResetPassword	http.Error	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B210	catalog-api/handlers/user_handler.go:694	LockAccount	http.Error	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B211	catalog-api/handlers/user_handler.go:738	UnlockAccount	http.Error	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)

#	File:line	Function	Sink	st	Expression	Class	Reason	HEAD
B212	catalog-api/internal/handlers/copy.go:249	CopyToStorage	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	binding/JSON-decode error; names Go struct fields/types, echoes malformed input	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B213	catalog-api/internal/handlers/localization_handlers.go:301	ImportConfiguration	h.sendError	http.StatusBadRequest	`F:fmt.Sprintf("Failed to import configuration: %v", err)`	LATENT	mixed: "failed to parse configuration JSON: %w" plus service/DB wrapping, sent via %v. UNROUTED: `LocalizationHandlers` is constructed only in `internal/handlers/localization_handlers_test.go`; no non-test registration (grep NewLocalizationHandlers / wizard/configuration in main.go: 0 hits)	CLOSED (err text no longer sent)
B214	catalog-api/internal/handlers/localization_handlers.go:365	EditConfiguration	h.sendError	http.StatusBadRequest	`F:fmt.Sprintf("Failed to edit configuration: %v", err)`	LATENT	mixed: service error wrapped via %v into a 400 body. UNROUTED: `LocalizationHandlers` is constructed only in `internal/handlers/localization_handlers_test.go`; no non-test registration (grep NewLocalizationHandlers / wizard/configuration in main.go: 0 hits)	CLOSED (err text no longer sent)
B215	catalog-api/internal/handlers/smb_discovery.go:86	DiscoverShares	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	binding/JSON-decode error; names Go struct fields/types, echoes malformed input	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B216	catalog-api/internal/handlers/smb_discovery.go:124	TestConnection	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	binding/JSON-decode error; names Go struct fields/types, echoes malformed input	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B217	catalog-api/internal/handlers/smb_discovery.go:176	BrowseShare	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	binding/JSON-decode error; names Go struct fields/types, echoes malformed input	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B218	catalog-api/internal/middleware/middleware.go:81	ErrorHandler	c.JSON	http.StatusBadRequest	`E:err.Error()`	LATENT	ErrorTypeBind branch; ErrorHandler is wired (main.go:1038) but no non-test code produces a gin bind error in c.Errors (no Bind/BindJSON/MustBindWith; ShouldBind* does not append)	CLOSED (err text no longer sent)
B219	catalog-api/internal/middleware/middleware.go:86	ErrorHandler	c.JSON	http.StatusInternalServerError	`E:err.Error()`	LATENT	ErrorTypePublic branch; no non-test code sets ErrorTypePublic (grep: 0 hits outside middleware.go)	CLOSED (err text no longer sent)
B220	catalog-api/internal/smartrules/smartrules.go:207	parseList	jsonErrText	-	`err`	VALIDATION	jsonErrText builds the authored ValidationError message from the caller own JSON decode error	unchanged; class unchanged (VALIDATION)
B221	catalog-api/main.go:1306	main	c.JSON	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)
B222	catalog-api/main.go:1337	main	c.JSON	http.StatusInternalServerError	`E:err.Error()`	LEAK	service/repo/DB error text sent verbatim on 5xx or generic status	CLOSED (err text no longer sent)

#	File:line	Function	Sink	st	Expression	Class	Reason	HEAD
B223	catalog-api/middleware/auth.go:167	RequireAuth	utils.SendErrorResponse	http.StatusUnauthorized	`err`	LEAK	JWT parse text (token, signing-method and claims detail) sent as details on 401	CLOSED-BY-HELPER (details no longer set; err logged)
B224	catalog-api/middleware/input_validation.go:360	InputValidation	c.JSON	http.StatusBadRequest	`E:err.Error()`	VALIDATION	deliberate validation text (may echo user input)	err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4)
B225	catalog-api/services/conversion_service.go:903	handleConversionError	s.notifyUser	-	`F:fmt.Sprintf("Conversion failed: %s", errorMsg)`	SAFE	notifyUser writes a debug log line and calls conversionNotifyHook (test-only setter; no production writer found); not a client sink	unchanged; class unchanged (SAFE)
B226	catalog-api/services/sync_service.go:1066	handleSyncError	s.notifyUser	-	`E:syncError.Error()`	SAFE	notifyUser writes a debug log line only (sync_service.go:1084-1093); not a client sink	unchanged; class unchanged (SAFE)

| H1 | catalog-api/internal/metrics/health.go:157 | checkDatabase | struct literal | - | `"Database ping failed: " + err.Error()` | LEAK | unauthenticated /health/deep body | CLOSED (fixed text, error logged) | | H2 | catalog-api/internal/monitoring/monitoring.go:251 | Check | map literal | - | `err.Error()` | LEAK (latent) | package unwired | STILL PRESENT (R1) |

Corrections of 2026-11-14 (PXF18-FU1 to FU11, FU6 excluded)

Each item was re-checked against main @ 9e624f09f (171 commits after 713a4f380; `git diff --stat 713a4f380 9e624f09f -- catalog-api`: 162 files changed) with the command shown; the scan counts were re-run on archives of 9ed6577b and 713a4f380 with the committed `docs/qa/err_error_class_scan/*.go.txt` sources. Line numbers of `main.go` and `handlers/scan_handler.go` moved between 713a4f380 and 9e624f09f (main.go: 345 changed lines, scan_handler.go: 12 added lines); rows above keep their 713a4f380 numbers unless a row says "at 9e624f09f".

FU	Status	What changed	Evidence (command and result)
FU1	corrected	S1 and R6: the WebSocket leg of a scan-failure `notification` is admin-only	`grep -n "audienceForType|AudienceAdmin" handlers/websocket_handler.go`: `audienceForType` at :74 returns `AudienceAuthenticated` only for `media_update` and `asset_update`, `AudienceAdmin` for everything else; `BroadcastToClients` (:604-606) and `BroadcastToClientsContext` (:641-653) skip non-admin connections; `git diff 713a4f380 9e624f09f -- catalog-api/handlers/websocket_handler.go` is empty
FU2	corrected	S5: field location and who can read it	`awk` over models: `ErrorMessage` at models/user.go:1108 is inside `type ConversionJob` (:1091); models/file.go:158 is inside `type ScanHistory` (:146). At 9e624f09f `authorizedJob` (conversion_service.go:1006) returns a job only to its owner or an administrator; `ListJobs` checks `PermissionConversionView` (conversion_handler.go:113) and `GetUserJobs` filters `user_id` (conversion_repository.go:278). At 713a4f380 `GetJob` (conversion_service.go:976-988) also admitted a non-owner holding `auth.PermissionViewMedia`; whether an ordinary role holds that permission was not checked: UNCONFIRMED
FU3	corrected	S3: route	`grep -n "storage/list" handlers/storage_routes.go`: :14 `api.GET("/storage/list/*path", auth.RequireAdmin(), h.ListStoragePath)`; `grep -rn "storage-roots/:id/list" --include=*.go` (non-test): 0 hits
FU4	reconciled	see "Reconciliation of the 310 sink-scan rows"	commands in that section
FU5	stated	flow-scan counts are after `sort -u`	`flows head/catalog-api | wc -l` = 365, `sort -u | wc -l` = 362; base 405 and 402; `sort | uniq -d` lists the same 3 rows at both bases
FU6	skipped	handled by another change, not edited here	(none)
FU7	corrected	R4 now lists conversion_handler.go:72-73 (B62)	`sed -n 70,76p handlers/conversion_handler.go`; `grep -rn ErrInvalidConversionRequest` (non-test): the only producer is conversion_service.go:83 `fmt.Errorf("%w: %s", ..., reason)` with `reason` from `conversionRequestProblem` ("a short client-safe reason", conversion_service.go:1448)

FU	Status	What changed	Evidence (command and result)
FU8	corrected	Limits bullets 2 and 3	fixture run in "Instrument blind spots measured 2026-11-14" below
FU9	added	"Checked-SAFE surfaces and scope exclusions" below	commands there
FU10	corrected	instrument numbering (listed 0, 1, 3, 2); S6 "three validators" is two of four; N4 closure row maps S6 to S9	`sed -n 943,975p services/configuration_service.go` (two validators return nil, two return fixed text); Class 4 table has rows S1 to S12
FU11	measured	reviewer fixtures R1 to R3 are missed by all three instruments; a partial typed scan found no live instance	"Instrument blind spots measured 2026-11-14" below

Post-merge reconciliation (PXF18DOC-F6, 2026-11-14, main @ 5a199fe36). Three statements above were true at their measurement base and changed afterwards: (1) R1 / H2 (`internal/monitoring/monitoring.go:251`, latent leak) is CLOSED by PXF23-F1 (the error is redacted; the package is still unimported, so the wiring decision stays open). (2) R3 and the challenge rows B20 to B24: `GET /api/v1/challenges`, `/challenges/:id` and `/challenges/results` are now admin-only through the group gate (PXF18-FU6), so the "gating of results was not established" remark is superseded; the response text there is operator-only. (3) The panic-recovery call `middleware.Recovery(logger)` is at main_router.go:89 (was :93 at 9e624f09f). Evidence: `grep -n Recovery catalog-api/main_router.go` at 5a199fe36; closures taken from the 2026-11-14 follow-up register, not re-derived here (UNCONFIRMED beyond that register).

Reconciliation of the 310 sink-scan rows (FU4)

Commands: build `astscan_sinks.go.txt`, run on `git archive 9ed6577b catalog-api`; `awk -F'\t' '$4 !~ /^(strings|sort|bytes)\./'` removes callee noise; compare the `file:line` set with the B<n> rows of the per-site table.

Step	Rows
Instrument 1 raw output, review base	432
After removing `strings.`, `sort.`, `bytes.` callees	310
of which under `catalog-api/challenges/` (harness, see R3)	60
outside `challenges/`	250
of which a second row at an already counted `file:line` (taint rows `T:` with a different expression at admin_handler.go:518 and :574, alias_routes.go:140, main.go:1013)	4
distinct `file:line` outside `challenges/`	246
of which in the per-site table (rows B1 to B226)	226
distinct `file:line` NOT in the table	20

310 - 226 = 84 = 60 (challenges/, see R3) + 4 (duplicate lines) + 20 (unlisted lines). The earlier heading "226 rows after removing `strings.*` callee noise" omitted the `sort.`/`bytes.` filter, the `challenges/` exclusion and the one-row-per-line rule; the control-needle bullet "310 and 250" is the first and second line of this table. The 20 unlisted lines, each read at the review base (<file>:<line> at 9ed6577b), are:

Lines	Reading	Disposition
scan_handler.go:377, 383, 400 (`ListStoragePath`, `services.ScrubErrorFor(err, root)` in a `c.JSON` body, kind `H:`)	the three call sites of Class 4 row S3 (713a4f380: :378, :384, :401); scan_handler.go:384 is the `Connect` failure branch (status `StatusBadGateway`), found by instrument 1 as `H:services.ScrubErrorFor(err, root)` at both bases, so it was not missed by the scan	S3 (LEAK to admins, partly scrubbed)
log_management_handler.go:48 (`writeLogServiceError`, `http.Error(w, msg, status)`, `T:status<-logServiceStatus(err)`)	`msg` is one of four fixed strings chosen by status ("Not found", "Access denied", "Unsupported format", "Internal server error"), the error value only selects the status	SAFE

Lines	Reading	Disposition
admin_handler.go:518, :574; alias_routes.go:140; conversion_handler.go:154; playlist_handler.go:64; scan_handler.go:301; service_handlers.go:360; internal/handlers/catalog.go:136; internal/services/media_recognition_service.go:349, :362; main.go:1013, :1732; models/user.go:81; repository/media_item_repository.go:534; services/log_management_service.go:325; services/pdf_text_isolation.go:369	identifiers that are not errors: `result`, `backups`, `out`, `jobs`, `len(playlists)`, `roots`, `len(favorites)`, `fileInfo`, `bestResult`, `fmt.Sprintf("ws://%s:%d/ws", host, port)`, `port`, `[]string(p)`, `v`, `entries`, `p`; no `error` value reaches the sink	noise (false positives of the BARE/`H:`/`F:` kinds)

Checked-SAFE surfaces and scope exclusions (FU9)

Surface	What was read	Result
panic recovery	exact command and control needle in the note "Panic-recovery grep (PXF18DOC-F4)" below: 0 hits. `middleware.Recovery(logger)` is installed at main_router.go:89 (was :93 at 9e624f09f, PXF18DOC-F6); `internal/middleware/recovery.go` logs a redacted panic and answers `c.AbortWithStatus(500)` with no body	SAFE
`assetEventBus`	main.go:770 (9e624f09f) forwards only `AssetReady`/`AssetFailed` events to `BroadcastToClients` with `type`, `action`, `asset_id`, `asset_type`, `entity_type`, `entity_id`; no error field	SAFE
`/logs` collection entries	`LogManagementHandler.GetLogEntries` (log_management_handler.go:238-269) requires `PermissionSystemAdmin` (:251) before returning entries; entries are the operator's own log data	ADMIN-ONLY by design, not a response-text leak
`ScanHistory.ErrorMessage` / `scan_history.error_message`	`grep -rn "INSERT INTO scan_history|UPDATE scan_history" --include=*.go` outside tests: only database/migrations_sqlite.go:183 (`UPDATE scan_history SET storage_root_id`); `AliasHandler.ScanHistory` (alias_routes.go:116) does not select the column; `StatsRepository.GetScanHistory` (repository/stats_repository.go:448, selecting `sh.error_message`) reads it into `internal/models/file.go:211` (`ScanHistoryItem`, type at :199; not `catalog-api/models/file.go:158`, which is the different struct `ScanHistory`) and `GET /stats/scans` serves it (corrected 2026-11-14, PXF18DOC-F3; measured with `sed -n 448p`, `grep -n ErrorMessage internal/models/file.go models/file.go` at 5a199fe36)	writer-less in catalog-api non-test Go (an external writer to the database is not excluded: UNCONFIRMED); nothing writes error text into it today
linked own-org submodule code	`grep -c '^replace' catalog-api/go.mod` = 23 (`../submodules/<name>`: assets, auth, cache, challenges, concurrency, config, containers, database, discovery, entities, event_bus, filesystem, lazy, media, memory, middleware, observability, rate_limiter, recovery, security, storage, streaming, watcher)	OUT OF SCOPE of this enumeration: the scans read the checked-in `catalog-api` tree only, the clones were made without submodule init; those modules were not scanned and no statement about them is made

Panic-recovery grep (PXF18DOC-F4). Run in `catalog-api/` at 5a199fe36; the result is 0 hits:

```
grep -rn "gin.Recovery\|gin.Default()" --include=*.go . | grep -v testdata | grep -v '_test.go:' | grep -v -E '^\S+:[0-9]+:\s*//'
```

Without the last two filters the same grep prints 33 lines. Control needle: without only the comment filter it prints exactly 2 non-test lines, both comments (main_router.go:87 and internal/middleware/recovery.go:19), so the instrument does see non-test matches and the 0 is real.

Instrument blind spots measured 2026-11-14 (FU8, FU11)

Six fixture functions were written to a scratch directory (never committed) and run through the three built instruments:

Fixture	Shape	Instrument 1 (sink)	Instrument 2 (census)	Instrument 3 (flow)
fixCause	`var cause error = do()`, `fmt.Sprintf("%v", cause)` inside `c.JSON`	found (`F:`)	not applicable (no `.Error()`)	found
fixE	`e := do()`, `fmt.Sprintf("%v", e)` inside `c.JSON`	missed	missed	missed
fixR1 (reviewer R1)	`_, boom := pair()`, `fmt.Sprintf("%v", boom)` inside `c.JSON`	missed	missed	missed
fixR1b	same, `boom.Error()`	found (`E:`)	found	found

Fixture	Shape	Instrument 1 (sink)	Instrument 2 (census)	Instrument 3 (flow)
fixR2 (reviewer R2)	`failure := do()`, `c.JSON(500, gin.H{"m": failure})`	missed	missed	missed
fixR3 (reviewer R3)	`c.JSON(500, gin.H{"m": h.Cause})`, `Cause error` struct field	missed	missed	missed

So bullets 2 and 3 of "Limits" were stale in two ways (the `cause` shape is found; DB round-trips are covered for S4 and S5) and the reviewer's three fixtures R1 to R3 are real blind spots of all three instruments. A typed scan was built to check the tree: a `go/packages` program (golang.org/x/tools v0.52.0, built offline in a scratch module; source committed as `docs/qa/err_error_class_scan/typed_scan_gopackages.go.txt`, PXF18DOC-F5; re-run 2026-11-14 on `git archive 713a4f380 catalog-api` with `GOPROXY=off`: 169 tab-separated rows, 168 `errname` + 1 `NOT-errname` (repository/asset_repository.go:202 `json.Unmarshal`), identical to the figures below; the scratch module's go.mod/go.sum are not committed, UNCONFIRMED that the committed file is byte-identical to the original run's source) that lists every expression whose type implements `error`, as an identifier or selector, in a sink-named call (same name list as instrument 1), skipping tests and `challenges/`. Run on an archive of 713a4f380 without submodules: 169 rows, 168 with an err-named identifier (already covered by instrument 1) and 1 with another name (repository/asset_repository.go:202, an argument of `json.Unmarshal`, not a client sink). The same program reported all six fixtures when they were placed inside the module. LIMIT: 18 packages failed to type-check because the submodules named in "Checked-SAFE surfaces" were absent (`could not import digital.vasic.*`), so the typed scan covered the rest of the tree only; the statement "no live instance" is therefore for the packages that type-checked and is otherwise UNCONFIRMED. The handler sweep `TestPXF23R5b_SourceSweep` (PXF23-F2) gained the same declared-type and error-synonym detection on 2026-11-14 (handlers/pxf23_f2_bare_err_test.go); it still does not see a struct field, an arbitrary-named multi-value result, a direct call result in a sink (`gin.H{"m": do()}`), or a variable / closure parameter of a concrete error-implementing type other than `error` (`var pe *os.PathError`); all five are pinned by `TestPXF23F2_KnownGaps` (PXF18DOC-F1). Known false-positive class (PXF18DOC-F2, measured 2026-11-14): the error-synonym name heuristic also fires on a NON-error value that carries a synonym name, e.g. `n, fail := lookup()` with a bool `fail` passed to a sink (1 report on the fixture); it over-reports (fails closed) and is pinned by `TestPXF23F2_KnownFalsePositives`; the bool `ok` negative control stays silent.

Limits of the instruments (measured, UNCONFIRMED where stated)

- Type information is heuristic: any zero-argument `.Error()` matches, including non-error types (`writer.Error()` of encoding/csv appears in the census; read, not a response); err-like identifiers are found by declared type `error` or a nil comparison, not by the type checker, so an `error` value obtained from a multi-value call and never compared with nil, a field of type `error`, or an interface method result is missed (round-2 blind spot, partly closed).
- Sink detection is by callee name; a response writer with a name outside the list is missed by instrument 1 and is caught only if instrument 2 sees an `.Error()` inside it. Measured 2026-11-14 with six fixtures (see "Corrections of 2026-11-14", FU8/FU11): a variable DECLARED `var cause error` and passed to `fmt.Sprintf` inside a sink IS found (kind `F:`, instrument 1 and 3); a short variable `e := do()` formatted with `%v`, a multi-value-call result with a non-err name (`_, boom := pair()`) formatted with `%v`, a non-err-named error passed bare to a sink (`failure`), and an error-typed struct field are NOT found by any instrument; `boom.Error()` IS found (instruments 1, 2, 3). Whether any such site exists in the tree: the partial typed scan below found none.
- Blind spots found by the round-2 review and closed only as far as stated: helper-in-sink (`H:`), non-err-named variable in `%v` (`F:`). Error text round-tripped through the database and served by a different route IS covered for the fields Class 4 traces (S4 `SyncSession.ErrorMessage`, S5 `ConversionJob.ErrorMessage`; `ScanHistory.ErrorMessage` has no writer, see "Checked-SAFE surfaces"); still NOT covered: any other database or file round-trip, event-bus consumers other than `EventBusBridge`, text built by `errors.Unwrap`/`errors.Join` chains and then formatted with `%s` of a non-error value, `fmt.Errorf` results returned to a handler that sends them (covered only when the handler's sink argument shows an err-like identifier), and routes registered through dynamic mux patterns. Class 4 is a trace of the fields instrument 3 surfaced, not a proof that no other field is served.
- Taint is flow-insensitive within a function (one false positive found and read, user_handler.go:122 at HEAD). Taint through struct fields, other functions or channels is not followed; the census catches the `.Error()` call itself but not where the stored text is later sent.
- BARE reports any err-named identifier argument to a sink-named callee, including calls whose helper never serialises it (that is why 111 SendErrorResponse rows are listed as CLOSED-BY-HELPER rather than as leaks).

- Producers: service and repository error strings were read only where a row says so (auth_service.go, configuration_service.go, error_reporting_service.go, local_policy.go, localization_service.go, playlist_service.go). Every other LEAK means "text can include internal detail", not a demonstrated leak.
- Not covered: Go modules other than catalog-api and OCU-CUDA-Sidecar (examples/, docs/qa/lib/ctrctl, catalog-api/tests/automation, tools/ fixtures were not scanned), JavaScript/TypeScript backends, generated code, and HTTP responses produced by third-party middleware. WebSocket (`handlers/websocket_handler.go`) and SSE (`log_management_handler.go`) code was grepped: error values there go to `h.logger` only (UNCONFIRMED beyond that grep).
- HEAD status was produced by matching each review-base row to the HEAD scan by (file, function, sink, kind, expression) and then reading the helper and the changed line for the rows that moved; line numbers shift, so the match ignores them.